12. Two short bar magnets have magnetic moments 1.20 Am^2 and 1.00 Am^2 respectively. They are kept on a horizontal table parallel to each other with their north poles pointing towards south. They have a common magnetic equator and are separated by a distance of 20.0 cm. The value of the resultant horizontal magnetic induction at the mid-point O of the line joining their centres is (Horizontal components of Earth's magnetic induction is 3.6×10^{-5} Wb m^{-2})

(NSEP 2000-2001)

(a) 3.60×10^{-5} Wb m^{-2}

(b) 3.5×10^{-5} Wb m^{-2}

(c) 2.56×10^{-4} Wb m^{-2}

(d) 2.2×10^{-4} Wb m^{-2}

13. The vertical component of Earth's magnetic field at a place is equal to the horizontal component. What is the value of angle of dip at this place?

(a) 30° (b) 45°

(c) 60° (d) 90°

14. A flat dielectric disc of radius R carries an excess charge on its surface. The surface charge density is σ. The disc rotates about an axis perpendicular to its plane passing through the centre with angular velocity ω. Find the magnitude of the torque on the disc if it is placed in a uniform magnetic field whose strength is B which is directed perpendicular to the axis of rotation

(a) $\frac{1}{4}\sigma\omega\pi BR$ (b) $\frac{1}{2}\sigma\omega\pi BR^2$

(c) $\frac{1}{4}\sigma\omega\pi BR^3$ (d) $\frac{1}{4}\sigma\omega\pi BR^4$

15. The potential energy of magnetic dipole whose dipole moment is $\vec{p}_m = (-0.5\hat{i} + 0.4\hat{j})$ Am^2 kept in uniform magnetic field $\vec{B} = 0.2\,\hat{i}$ T

(a) –0.1 J (b) –0.8 J

(c) 0.1 J (d) 0.8 J

Answers

1) a	**2)** d	**3)** c	**4)** b	**5)** b
6) a	**7)** b	**8)** a	**9)** b	**10)** c
11) c	**12)** c	**13)** b	**14)** d	**15)** c

II Short answer questions:

1. What is magnetic field?

2. Define magnetic flux.

3. Define magnetic dipole moment.

4. State Coulomb's inverse law.

5. What is magnetic susceptibility?

6. State Biot-Savart's law.

7. What is magnetic permeability?

8. State Ampere's circuital law.

9. Compare dia, para and ferro-magnetism.

10. What is meant by hysteresis?

11. Define magnetic declination and inclination.

12. What is resonance condition in cyclotron?

13. Define ampere.

14. State Fleming's left hand rule.

15. Is an ammeter connected in series or parallel in a circuit? Why?

16. Explain the concept of velocity selector.

17. Why is the path of a charged particle not a circle when its velocity is not perpendicular to the magnetic field?

18. Give the properties of dia / para / ferromagnetic materials.

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